

Hany Ehab Elashry

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PROFILE

Machine Learning Engineer specializing in from-scratch algorithmic derivation, production-grade data pipelines, and full end-to-end ML system delivery. Experienced across the complete lifecycle: relational schema design, model training, and deploying live services (FastAPI + PostgreSQL backends paired with interactive Streamlit dashboards). Seeking a junior/internship role applying strong Python, SQL, and applied ML skills to real, data-heavy engineering problems.

EDUCATION

B.Sc. in Navigation & Space Technology *Beni Suef University | Expected: July 2028*

GPA: 3.6/4.0 | Focus: Space Systems, Orbital Mechanics, Advanced Physics, Mathematics

Data Engineering Track *Digital Egypt Pioneers Initiative (DEPI) | In Progress*

Python & data manipulation (NumPy, Pandas), SQL and database fundamentals, web scraping, ETL pipelines, and data warehousing.

TECHNICAL SKILLS

Languages: Python (Advanced), C++, SQL, MATLAB, JavaScript, HTML/CSS.

Data & ML: NumPy (Expert), Pandas, Scikit-Learn, SciPy, XGBoost, Matplotlib, Plotly.

Data Engineering: PostgreSQL, MySQL, MongoDB/NoSQL, SQLAlchemy (declarative ORM, relational schema design), ETL pipeline architecture.

Backend & Deployment: FastAPI, REST API design, Streamlit, Railway, Streamlit Community Cloud, environment-based configuration.

Domain Tech: Orbital Mechanics, FMCW Radar Signal Processing, LVLH Coordinate Systems.

Tools: Git, GitHub, DBeaver, Jupyter Ecosystem, VS Code.

Currently Learning: PyTorch & Deep Learning (intermediate — halfway through applied coursework, including CNNs and custom-dataset training pipelines).

EXPERIENCE & LEADERSHIP

Team Lead, Sky-Spy Project *NASA Space Apps Hackathon | Sep 2025*

- Directed a cross-functional team to 3rd place by architecting an exoplanet classification system, from raw Kepler mission data to a fully deployed, publicly accessible service.
- Built a tuned XGBoost classification pipeline (5-fold grid search) to isolate confirmed exoplanets from false-positive transit signals.
- Deployed the system as two independent live services — a FastAPI prediction backend and an interactive Streamlit dashboard — hosted on separate cloud platforms.

Technical Lead, FMCW Radar Simulation Engine *Independent Team Project | Ongoing*

- Leading the DSP pipeline for a 4-person team building a full FMCW radar simulation environment, targeting realistic scenario generation (structured noise and synthetic targets) alongside CFAR detection and ML/CNN-based target classification.
- Completed the core DSP processing stage, extracting a full range-velocity-angle data cube from simulated radar returns; CFAR detection is the current active development stage.

KEY PROJECTS

TicketSense: Self-Built ETL Pipeline for Support Ticket Data *Python, PostgreSQL, SQLAlchemy*

- Designed and implemented a normalized relational schema (customers, tickets) from a flat, denormalized dataset — resolving real data-integrity issues including 441 duplicate customer records identified during EDA.
- Built a modular ETL pipeline (extract, transform, load) using SQLAlchemy's declarative ORM with typed models and foreign-key relationships, independently identifying and correcting a data quality defect (unresolved template placeholders in raw text fields) not flagged in the source dataset.

Kessler-Shield: Orbital Collision Early-Warning System *Python, XGBoost, FastAPI, Streamlit*

- Built a custom XGBoost classification pipeline predicting orbital collision risk from telemetry data, natively resolving a ~1:1300 class imbalance without synthetic oversampling, using precision-recall-driven threshold tuning.
- Diagnosed and resolved a production model-serving defect caused by a library version mismatch between the training and deployment environments, restoring correct inference behavior on the live service.
- Deployed as a full two-service architecture — an independent FastAPI prediction API and a Streamlit dashboard — running on separate cloud hosts and communicating over a REST interface.

The ML Blueprint: Mathematically Applied Reference *Python, NumPy*

- Derived foundational ML algorithms from first principles in NumPy, including a from-scratch neural network with manually implemented forward and backward passes.

LANGUAGES

Arabic (Native), English (C1 — Professional Fluency)
